# UNIT 3

## FUNCTIONS
## DATA STRUCTURE
## OOP

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# FUNCTION

- A function is a block of organized and reusable program code that performs a single, specific and well-defined task.

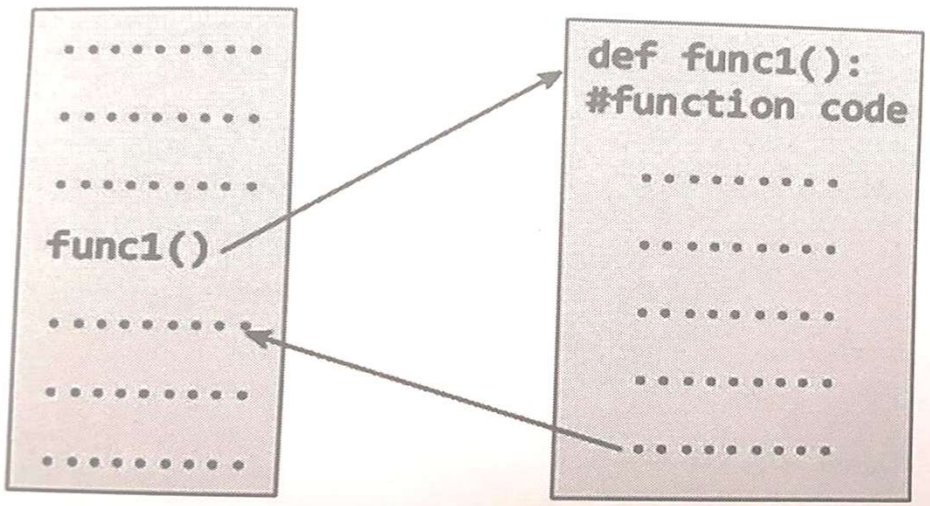



**Figure 3.1** Calling Function func1()

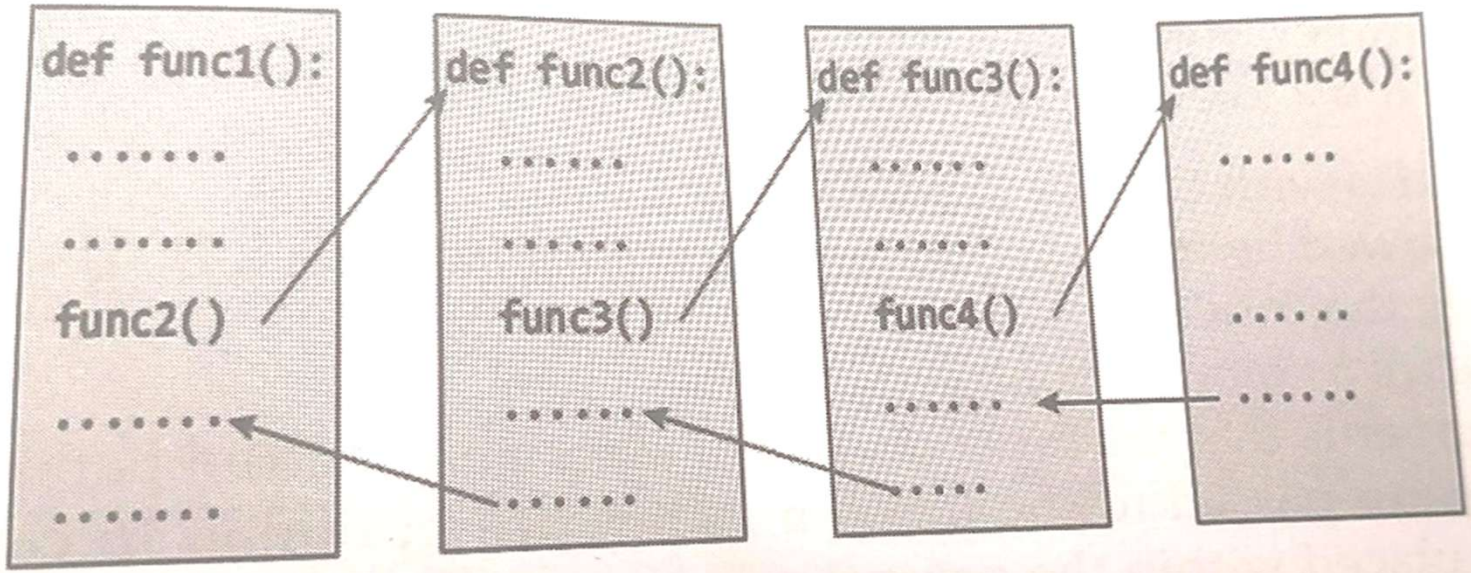



Figure 3.2 Function(s) Calling Another Function

Here func1() calls another function named func2().
func1() is known as ***calling function.***
func2() is known as ***called function.***

When the compiler encounters a function call, instead of executing the next statement in calling function, the control jumps to the statements that are a part of the called function.

# Pointers to remind:

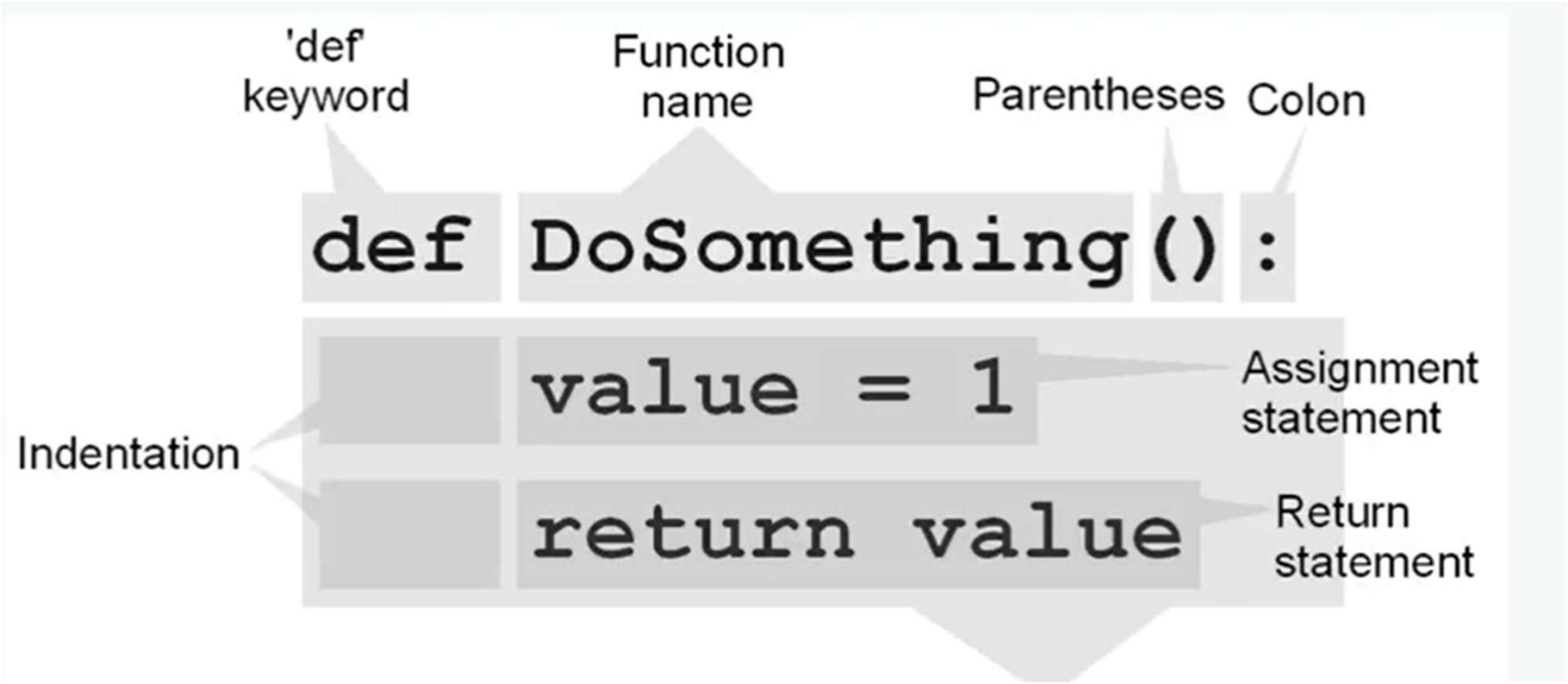
- A function, ***f*** that uses another function ***g,*** is known as calling function and g is known as the called function.
- The inputs that the function takes are known as *arguments/parameters*.
- When a called fn returns some result back to the calling fn, its called *return* that **result.**
- If the called fn accepts arguments, the calling fn will pass parameters otherwise not.

# Function Declaration & Definition

Function declaration : A declaration statement that identifies a function with its name, a list of arguments that it accepts and the type of data it returns.

Function definition : It consists of a function header that identifies the function, followed by the body of the function containing the executable code for that function.

# PYTHON FUCTION




FUNCTION BODY

# Define a function – User defined

- Function blocks starts with the keyword **def.**
- Keyword followed by the function name and (). The fn name is used to uniquely identify the function.
- After (), place colon(:)
- Parameters are in parentheses. Values are passed through them and are optional. If not used, the parentheses are empty.
- The code block within the function id properly indented to form the block code.
- A function's return statement is optional. If presents, it sends an [expression] back to the caller. A return statement with no argument is equivalent to return None.

Program that adds two numbers using function

```
def add(x, y):     # function to add two numbers
        return x+y
a = 20
b = 10
print(add(a,b))
```

OUTPUT:
30

When a function is defined, space is allocated for that function in the memory.
A function definition comprises of 2 parts:

- Function Header
- Function Body

The syntax of a function definition can be given as:

```
def function_name(variable1, variable2,…):
        documentation string
        statement block
        return [expression]
```

**Function Header**

**Function Body**

Program to write a function that displays a string repeatedly for 4 times

```
def fun():
        for i in range(4):
                print("Hello World")
func()    # function call
```

OUTPUT:

```
Hello World
Hello World
Hello World
Hello World
```